

INTERNET OF THINGS (IoT)

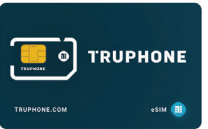
Wireless Technologies Used in IoT

Technologies	Frequency	Data Rate (up to)	Range (up to)	Power	Cost
LTE-M/ NB-IoT	Cellular Bands	~ 1 Mbps	Several km	Low	Low
LTE CAT-1	Cellular Bands	10 Mbps	Several km	High	Medium
3G	Cellular Bands	20 Mbps	Several km	High	High
2G	Cellular Bands	384 Kbps	Several km	High	Medium
LoRa	Sub 1 GHz	< 100 Kbps	Several km	Low	Low
SIGFOX	Sub 1 GHz	< 50 Kbps	Several km	Low	Low
Wi-Fi	2.4, 5 GHz	1 - 100+ Mbps	100 Meters	High	Low
Bluetooth	2.4 GHz	~ 1-2 Mbps	100 Meters	Low	Low
802.15.4/ Zigbee/Thread	2.4 GHz	250 Kbps	10 Meters	Low	Low
Z-Wave	Sub 1 GHz	< 100 Kbps	100 Meters	Low	Low

IoT Cellular Data Plans

Digi-Key has partnered with several tier-1 wireless carriers to offer cellular connectivity services. Our goal is to provide a path to create end-to-end IoT solutions. These data plans complement Digi-Key's offering of cellular modules, modems, and gateways. Now Digi-Key offers you a fast and simple way to get connected with these devices.

SIM Cards

Verizon	AT&T	T-Mobile	Truphone
			
Triple 2FF/3FF/4FF 1998-1012-ND CAT-M 2FF/3FF/4FF 1998-251-ND	Triple 2FF/3FF/4FF 1998-1011-ND	Triple 2FF/3FF/4FF 1998-1003-ND	Triple 2FF/3FF/4FF 2322-SIM-E-I03-TRI-2-ND Cat-M 2FF/3FF/4FF 2322-SIM-S-I03-TRI-2-LP-ND
These SIM cards are universal and come in a breakaway package. Each SIM card comes in multiple form factors: Mini (2FF), Micro (3FF) and Nano (4FF).			

eSIM Cards

Have you considered using an eSIM card? eSIM is short for "embedded SIM" card. These SIMs are soldered on the board, saving space and cost vs. using the larger plastic SIMs and connectors. eUICC (embedded Universal Integrated Circuit Card) eSIMs can give flexibility in choosing which carrier profiles you want to use and the ability to switch on the fly to access lower rates, better coverage, other countries, etc. eSIMs are one of the smallest form factors for SIM cards, making them ideal for use in electronics that have small profiles. eSIM technology opens up new possibilities for machine-to-machine (M2M), Internet of Things (IoT), and consumer devices. It offers new ways of managing and deploying connectivity, without the hassle of traditional SIM cards.

For more information, visit [DIGIKEY.COM/NOTEBOOK](https://www.digikey.com/notebook)